

Empirical IO

Chapter 3 Problem Set

Production and Cost Function Estimation

With detailed worked answers

What this PDF includes. This set turns the chapter into a study-ready workbook. It covers the logic of production-function estimation, endogeneity and selection bias, the Olley-Pakes and Levinsohn-Petrin approaches, the ACF collinearity critique, share-regression intuition, cost duality, translog cost functions, and revealed-preference cost identification under monopoly and oligopoly.

Suggested use: Try each question on your own first, then compare with the worked answer. The questions are designed to be useful for revision, problem classes, and exam preparation.

Based on Chapter 3 course content

Prepared as a worked problem set

Orientation

Chapter map.

- **Part I** covers the motivation for production-function work and the main bias problems.
- **Part II** covers proxy-variable estimators, identification issues, and modern critiques.
- **Part III** covers cost-function estimation and revealed-preference approaches.

Notation used throughout. Lowercase variables are logs unless otherwise stated.

Symbol	Meaning
y_{jt}	log output of firm j at time t
l_{jt}	log labor input
k_{jt}	log capital input
i_{jt}	log investment
m_{jt}	log intermediate or material input
ω_{jt}	anticipated productivity shock observed by the firm
η_{jt}	ex-post or transitory shock not anticipated when inputs are chosen
ν_{jt}	innovation in the productivity process
χ_{jt}	indicator that the firm continues in the market
P_{jt}	continuation or survival probability (propensity score)

Part I. Foundations and Sources of Bias

Problem 1. Why estimate production and cost functions?

Question. Explain why production and cost functions are central objects in empirical industrial organization. Distinguish among production, cost, supply (or pricing), and profit functions, and explain why empirical researchers often want parameters that are both primitive and as reduced-form as possible.

Detailed answer. Production functions describe how inputs map into output. Estimating them lets the researcher separate the contribution of observed inputs from the residual component usually called *productivity*. That is why the chapter treats production-function estimation as the starting point for questions such as: What drives firm productivity? How do deregulation, trade, R&D, or management practices affect performance?

The objects are nested in the following sense:

- A **production function** maps inputs into output.
- Adding a factor-market environment and cost minimization yields a **cost function**.
- Adding product-market structure and profit maximization yields a **supply or pricing rule**.
- Combining pricing and cost implies a **profit function**.

Researchers want parameters that are **primitive enough to be policy invariant**. For example, if a policy experiment changes the factor market, a previously estimated cost function may not remain stable, while a production function can still be a meaningful primitive object. At the same time, among the parameters that are primitive enough, researchers usually prefer the most **reduced-form** representation that is still valid, because reduced-form objects are often more robust to misspecification. A flexible nonparametric cost function, for instance, may still be useful even when firms do not literally solve the exact structural cost-minimization problem imposed by the econometrician.

Problem 2. Cobb-Douglas as a benchmark model

Question. Suppose the production function is

$$Y_{jt} = A_{jt} L_{jt}^{\beta_l} K_{jt}^{\beta_k}.$$

Derive the log-linear specification used in the chapter and interpret each parameter.

Detailed answer. Taking logs gives

$$y_{jt} = \beta_0 + \beta_l l_{jt} + \beta_k k_{jt} + \epsilon_{jt},$$

where

$$\ln A_{jt} = \beta_0 + \epsilon_{jt}.$$

The interpretation is straightforward:

- β_l is the output elasticity with respect to labor.
- β_k is the output elasticity with respect to capital.
- β_0 is a common intercept term, which can be interpreted as the average log productivity level.
- ϵ_{jt} captures firm-time specific unobserved heterogeneity: better machines, worker skill, managerial quality, unobserved technology, and similar factors.

This specification is useful because it converts the problem into a linear regression model. However, the chapter immediately warns that the linear form alone does *not* justify OLS, because the regressors are typically chosen by firms that know more about productivity than the econometrician does.

A final useful implication is that the sum $\beta_l + \beta_k$ measures returns to scale in the Cobb-Douglas case. If the sum equals one, the technology has constant returns to scale; if it is greater than one, returns are increasing; if it is less than one, returns are decreasing.

Problem 3. Why OLS is biased: endogeneity of inputs

Question. Explain the endogeneity problem in production-function estimation. Why does OLS typically overstate the labor coefficient in the benchmark static case?

Detailed answer. The main problem is that firms choose inputs after observing information about their own productivity. If the manager knows that current productivity is high, she has an incentive to use more inputs, especially flexible inputs such as labor. Then the chosen input is positively correlated with the unobserved productivity term in the regression.

In the benchmark static case with labor chosen given capital, the firm's labor demand takes the form

$$L_{jt} = \left[\frac{p_{jt}}{w_{jt}} \beta_l \exp(\beta_0 + \epsilon_{jt}) K_{jt}^{\beta_k} \right]^{1/(1-\beta_l)}.$$

A higher realization of ϵ_{jt} raises desired labor. Therefore,

$$\text{Cov}(l_{jt}, \epsilon_{jt}) > 0.$$

If the econometrician omits ϵ_{jt} and runs OLS, part of the increase in output caused by the favorable productivity shock is incorrectly attributed to labor. In that benchmark case, the estimated labor elasticity is biased upward. More generally, the sign and magnitude of the bias can depend on timing, adjustment frictions, and which input is being considered, but the core lesson is always the same: input choices are endogenous because firms observe more about their productivity than the econometrician does.

Problem 4. Selection bias from firm exit

Question. Even if capital is predetermined, why can the capital coefficient still be biased in a sample of surviving firms only? Give the selection-bias intuition.

Detailed answer. Firms do not remain in the market at random. Low-productivity firms are more likely to exit, but firms with large capital stocks may still survive even after drawing relatively poor productivity realizations. As a result, the sample of observed firms is selected on the basis of both productivity and capital.

The key implication is that, in the survivor sample,

$$E[\omega_{jt} \mid k_{jt}, \chi_{jt} = 1] \neq E[\omega_{jt} \mid k_{jt}],$$

so capital becomes correlated with the unobserved productivity term *conditional on survival*.

The problem does not require capital choice itself to respond to current productivity. It is enough that survival depends on productivity and capital jointly. This is why the chapter emphasizes that selection bias is conceptually distinct from the usual input-endogeneity problem. Endogeneity comes from firms choosing inputs after observing productivity. Selection comes from the fact that we only observe firms that survive the productivity process.

The sign of the induced bias in the capital coefficient is not mechanically fixed in every model, but the presence of correlation is enough to invalidate naive estimators.

Problem 5. What panel data can and cannot fix

Question. Suppose

$$\epsilon_{jt} = \mu_j + \eta_{jt},$$

where μ_j is a firm fixed effect and η_{jt} is mean independent of input choices up to period t . Derive the first-difference equation and explain both the attraction and the weakness of this approach.

Detailed answer. Under the maintained assumption,

$$y_{jt} = \beta_0 + \beta_l l_{jt} + \beta_k k_{jt} + \mu_j + \eta_{jt}.$$

Differencing between periods t and $t - 1$ removes the time-invariant component:

$$y_{jt} - y_{j,t-1} = \beta_l(l_{jt} - l_{j,t-1}) + \beta_k(k_{jt} - k_{j,t-1}) + (\eta_{jt} - \eta_{j,t-1}).$$

This is attractive because it eliminates unobserved firm heterogeneity that does not vary over time. If η_{jt} is orthogonal to the differenced regressors, then fixed effects or first differences can identify the elasticities.

The weakness is that the assumption is quite restrictive. It requires the persistent component of productivity to be time invariant at the firm level. In many empirical settings, productivity is serially correlated and evolves over time, so differencing does not solve the fundamental simultaneity problem.

A second weakness is measurement error. First differencing reduces signal but often leaves a large portion of the noise. That can make fixed-effects estimators more biased than OLS when inputs are measured with error.

Problem 6. IV methods and first-order conditions

Question. Discuss two common instrument strategies mentioned in the chapter: input prices and lagged inputs. Under what conditions can they work, and what are their main failure modes? What extra structure is required by first-order-condition based approaches?

Detailed answer. The chapter highlights two candidate instrument classes.

1. Input prices. These can work if they shift input demand but are orthogonal to the firm-specific productivity shock. That logic is strongest when factor markets are competitive and the shock is idiosyncratic to the firm. In practice, however, input prices often have very little cross-sectional variation, and when they do vary, part of that variation may reflect unobserved input quality. Then the exclusion restriction becomes doubtful.

2. Lagged inputs. These can be valid if current productivity shocks are not serially correlated and if adjustment frictions make lagged inputs predictive of current inputs. But if productivity follows, say, an AR(1) process,

$$\epsilon_{jt} = \alpha\epsilon_{j,t-1} + \nu_{jt},$$

then all sufficiently recent lags remain correlated with the current error term through the persistent productivity process. In that case lagged inputs are invalid instruments.

First-order-condition approaches use optimality conditions directly as moment conditions. That can be powerful, but it requires extra economic structure: the researcher must specify the relevant factor-market environment and the firm's optimization problem. In other words, one needs assumptions on market structure, adjustment costs, timing, and whether the firm is minimizing cost or maximizing profit.

Part II. Proxy-Variable Estimators and Identification Issues

Problem 7. The logic of the Olley-Pakes assumptions

Question. State the main assumptions behind the Olley-Pakes (OP) approach and explain the role played by each one.

Detailed answer. OP decomposes the composite error term into

$$\epsilon_{jt} = \omega_{jt} + \eta_{jt},$$

where ω_{jt} is an anticipated productivity shock observed by the firm when it chooses inputs, and η_{jt} is an ex-post shock that is mean independent of the firm's information set.

The main assumptions are:

- **Information set assumption.** The firm's information set at time t contains current and past productivity shocks, but not future shocks. The transitory shock satisfies

$$E[\eta_{jt} \mid I_{jt}] = 0.$$

This creates valid orthogonality conditions for the first stage.

- **First-order Markov productivity.** Productivity evolves according to

$$p(\omega_{j,t+1} \mid I_{jt}) = p(\omega_{j,t+1} \mid \omega_{jt}),$$

or equivalently

$$\omega_{jt} = E[\omega_{jt} \mid \omega_{j,t-1}] + \nu_{jt}.$$

This reduces the state dimension and underpins the second-stage control function.

- **Timing of inputs.** Capital is accumulated through past investment,

$$k_{jt} = \kappa(k_{j,t-1}, i_{j,t-1}),$$

while labor is chosen contemporaneously. This distinguishes slow-moving capital from flexible inputs.

- **Scalar unobservable.** Investment follows a policy rule

$$i_{jt} = f_t(k_{jt}, \omega_{jt}),$$

meaning that only one latent state, ω_{jt} , drives the policy once capital is given.

- **Strict monotonicity.** The investment rule is strictly increasing in ω_{jt} . This lets the econometrician invert the policy function and recover a proxy for productivity from observed (k_{jt}, i_{jt}) .

Economically, OP succeeds because it translates a dynamic model of firm behavior into a control function that absorbs the anticipated productivity shock.

Problem 8. The first step of Olley-Pakes

Question. Show how OP uses investment as a proxy for productivity in the first step. What is the function $\phi(k_{jt}, i_{jt})$, and under what condition is β_l identified?

Detailed answer. Under the scalar unobservable and monotonicity assumptions,

$$i_{jt} = f_t(k_{jt}, \omega_{jt}) \implies \omega_{jt} = h(k_{jt}, i_{jt}).$$

Start from the production function

$$y_{jt} = \beta_0 + \beta_l l_{jt} + \beta_k k_{jt} + \omega_{jt} + \eta_{jt}.$$

Substituting the inverse policy function gives

$$y_{jt} = \beta_l l_{jt} + \underbrace{[\beta_0 + \beta_k k_{jt} + h(k_{jt}, i_{jt})]}_{\phi(k_{jt}, i_{jt})} + \eta_{jt}.$$

So the first-stage equation is

$$y_{jt} = \beta_l l_{jt} + \phi(k_{jt}, i_{jt}) + \eta_{jt}.$$

The function $\phi(\cdot)$ is an unknown control function that bundles together the intercept, the direct capital term, and the anticipated productivity shock as revealed by investment. Because η_{jt} is ex-post and mean independent of the firm's information set, one can estimate β_l and ϕ from the moment condition

$$E[\eta_{jt} \mid l_{jt}, k_{jt}, i_{jt}] = 0.$$

However, this only works if there is enough independent variation in labor once the nonparametric function of (k_{jt}, i_{jt}) has been partialled out. That “enough variation” condition turns out to be delicate and is exactly what later motivates the ACF critique.

Problem 9. The second step of Olley-Pakes and recovery of productivity

Question. Once the first stage has estimated β_l and ϕ , how does OP recover β_0 , β_k , and firm-level productivity?

Detailed answer. By definition,

$$\omega_{jt} = \phi(k_{jt}, i_{jt}) - \beta_0 - \beta_k k_{jt}.$$

Now use the Markov law of motion

$$\omega_{jt} = g(\omega_{j,t-1}) + \nu_{jt},$$

where ν_{jt} is the productivity innovation. Substituting into the production function gives

$$y_{jt} - \hat{\beta}_l l_{jt} = \beta_0 + \beta_k k_{jt} + g\left(\hat{\phi}(k_{j,t-1}, i_{j,t-1}) - \beta_0 - \beta_k k_{j,t-1}\right) + \tilde{\eta}_{jt},$$

where $\tilde{\eta}_{jt}$ collects the innovation, the transitory shock, and first-stage estimation error.

The second step identifies β_0 , β_k , and the unknown function $g(\cdot)$ using moment conditions of the form

$$E[\nu_{jt} + \eta_{jt} \mid k_{jt}, k_{j,t-1}, i_{j,t-1}] = 0.$$

In practice, g is approximated with polynomials or splines.

Once $\hat{\beta}_0$ and $\hat{\beta}_k$ are available, firm-level productivity is recovered as

$$\hat{\omega}_{jt} = \hat{\phi}(k_{jt}, i_{jt}) - \hat{\beta}_0 - \hat{\beta}_k k_{jt}.$$

So the OP estimator does more than estimate elasticities. It also produces a firm-level estimate of anticipated productivity, which is often the final object of economic interest.

Problem 10. Why selection correction enters only later in OP

Question. The chapter says that exit does not affect the first step but does bias the second step. Explain why, and summarize how the OP correction works.

Detailed answer. In the first step,

$$y_{jt} = \beta_l l_{jt} + \phi(k_{jt}, i_{jt}) + \eta_{jt}.$$

Exit decisions depend on anticipated productivity ω_{jt} , not on the ex-post shock η_{jt} . Therefore, even in the selected sample of surviving firms,

$$E[\eta_{jt} \mid l_{jt}, k_{jt}, i_{jt}, \chi_{jt} = 1] = 0,$$

so the first-stage orthogonality condition remains valid.

The second step is different because it relies on the innovation in the productivity process. Survival depends on the current productivity realization, so among continuing firms,

$$E[\nu_{jt} \mid k_{jt}, k_{j,t-1}, i_{j,t-1}, \chi_{jt} = 1] \neq 0.$$

That is the source of the bias.

OP corrects this with a selection control. If continuation is optimal whenever productivity exceeds a threshold,

$$\chi_{jt} = 1\{\omega_{jt} \geq \bar{\omega}(k_{jt})\},$$

then the continuation probability

$$P_{jt} = \Pr(\chi_{jt} = 1 \mid i_{j,t-1}, k_{j,t-1})$$

contains information about the selection rule. The threshold can be represented as a function of the propensity score and lagged productivity, so the second-stage control function becomes

$$\tilde{g}(\omega_{j,t-1}, P_{jt}),$$

rather than just $g(\omega_{j,t-1})$.

Operationally, OP adds a **zero step**: estimate the survival probability, then insert the estimated propensity score into the second-stage regression.

Problem 11. The zero-investment problem and why Levinsohn-Petrin helps

Question. Why does frequent zero investment create a problem for OP? Compare the two fixes discussed in the chapter.

Detailed answer. OP depends on being able to invert the investment rule:

$$i_{jt} = f_t(k_{jt}, \omega_{jt}) \implies \omega_{jt} = h(k_{jt}, i_{jt}).$$

If many firms choose $i_{jt} = 0$, the policy function is flat over a region of the state space, so the mapping from productivity to investment is no longer one-to-one. Then the control function breaks down.

The chapter gives two responses.

Fix 1: drop zero-investment observations. If one restricts attention to observations with strictly positive investment, invertibility can be restored on that selected sample. The estimator need not become inconsistent, because the next-period innovation ν_{jt} is independent of events known up to period $t - 1$. The downside is potentially severe information loss.

Fix 2: use another proxy. Levinsohn and Petrin (LP) replace investment with intermediate inputs such as materials, which typically vary more smoothly and are rarely zero. If materials obey

$$m_{jt} = m(k_{jt}, \omega_{jt}) \implies \omega_{jt} = h(m_{jt}, k_{jt}),$$

they can serve as an alternative proxy for anticipated productivity.

In short, LP keeps the proxy-variable logic of OP but chooses a proxy that is empirically more usable in many plant-level data sets.

Problem 12. The LP estimator in a gross-output production function

Question. Suppose output depends on materials directly:

$$y_{jt} = \beta_0 + \beta_l l_{jt} + \beta_k k_{jt} + \beta_m m_{jt} + \omega_{jt} + \eta_{jt}.$$

Explain the LP first and second steps and clarify why current materials cannot be used in the second-step conditioning set.

Detailed answer. If materials are a monotone function of current capital and anticipated productivity, then

$$\omega_{jt} = h(m_{jt}, k_{jt}).$$

Substituting into the production function yields the first-stage equation

$$y_{jt} = \beta_l l_{jt} + \phi(m_{jt}, k_{jt}) + \eta_{jt},$$

where

$$\phi(m_{jt}, k_{jt}) = \beta_0 + \beta_k k_{jt} + \beta_m m_{jt} + h(m_{jt}, k_{jt}).$$

The first stage identifies β_l and the unknown control function ϕ .

The second stage uses the Markov law for productivity:

$$y_{jt} - \beta_l l_{jt} = \beta_0 + \beta_k k_{jt} + \beta_m m_{jt} + g(\phi(m_{j,t-1}, k_{j,t-1}) - \beta_0 - \beta_k k_{j,t-1} - \beta_m m_{j,t-1}) + \nu_{jt} + \eta_{jt}.$$

Crucially, the moment conditions cannot condition on *current* materials m_{jt} , because materials respond to the current anticipated productivity shock and are therefore correlated with the second-stage innovation term. Instead, LP conditions on lagged state variables such as $k_{jt}, m_{j,t-1}, k_{j,t-1}$. The identification of β_m comes from the appearance of $m_{j,t-1}$ in the lagged control function.

So LP solves the zero-investment problem, but it requires extra care because the proxy input is also a direct input in the production function.

Problem 13. Why one-step GMM can improve on two-step estimation

Question. What is the main idea behind the one-step estimator associated with Wooldridge (2009), and why can it be more efficient than the original two-step procedures?

Detailed answer. The one-step approach keeps the OP or LP logic but estimates all parameters jointly using a single GMM problem. Instead of first estimating a control function and then plugging it into a second-stage regression, the researcher stacks the moment conditions from both stages.

For the OP setup, the first set of moments comes from the ex-post shock:

$$E[y_{jt} - \beta_0 - \beta_l l_{jt} - \beta_k k_{jt} - \omega(k_{jt}, i_{jt}) \mid l_{jt}, k_{jt}, i_{jt}] = 0.$$

The second set comes from the law of motion for productivity:

$$E[y_{jt} - \beta_0 - \beta_l l_{jt} - \beta_k k_{jt} - g(\omega(k_{j,t-1}, i_{j,t-1})) \mid k_{jt}, i_{j,t-1}, l_{j,t-1}] = 0.$$

Joint estimation can be more efficient for two reasons:

- It uses all valid moments at once instead of treating first-step estimates as fixed inputs.
- It propagates first-stage uncertainty correctly inside the final estimator rather than only after the fact.

The trade-off is computational. One-step GMM is heavier numerically and can be more sensitive to the choice of basis functions, instruments, and weighting matrices. But when implemented well, it often extracts more information from the same data.

Problem 14. The ACF collinearity critique

Question. Akerberg, Caves, and Frazer argue that labor may not be identified in the standard OP/LP first stage. Explain the collinearity problem clearly.

Detailed answer. The issue begins with the firm's labor policy. If labor is chosen after observing the same state variables that determine the proxy input, then labor itself is a function of those same state variables:

$$l_{jt} = l(k_{jt}, \omega_{jt}).$$

But in OP,

$$\omega_{jt} = h(i_{jt}, k_{jt}),$$

so labor can be rewritten as

$$l_{jt} = \tilde{l}(i_{jt}, k_{jt}).$$

Now consider the first-stage regression:

$$y_{jt} = \beta_0 + \beta_l l_{jt} + \phi(i_{jt}, k_{jt}) + \eta_{jt}.$$

If l_{jt} is already determined by (i_{jt}, k_{jt}) , then the labor regressor contains no independent variation beyond what is already absorbed by the nonparametric function $\phi(i_{jt}, k_{jt})$. In effect, the model becomes

$$y_{jt} = \tilde{\phi}(i_{jt}, k_{jt}) + \eta_{jt},$$

so β_l cannot be separated out.

The problem does not disappear automatically in the second stage, because lagged labor may also be a deterministic function of lagged proxy variables and lagged capital. Then there is still no remaining variation to identify the labor elasticity. The ACF critique therefore says that labor identification in the original proxy-variable estimators relies on stronger timing assumptions than was initially recognized.

Problem 15. How to respond to the collinearity problem

Question. Summarize the two broad responses in the chapter: modified timing assumptions and share regression. What are the strengths and limitations of each?

Detailed answer. The first response is to change the timing assumptions so that the proxy choice can depend on labor, rather than labor being determined by the same information as the proxy. In the Akerberg-style timing story, labor is chosen after an unexpected shock but before materials or investment, so the proxy policy can take the form

$$m_{jt} = m(k_{jt}, \omega_{jt}, l_{jt}).$$

Then the first stage becomes

$$y_{jt} = \psi(k_{jt}, m_{jt}, l_{jt}) + \eta_{jt},$$

and identification of labor can be recovered in the second stage using lagged variables.

The limitation is important: for this timing fix to work cleanly, materials generally cannot also enter the production function directly. Otherwise the current and lagged materials coefficients create a new identification problem. That is why the chapter says the approach is much more natural for a **value-added** specification than for a gross-output one.

The second response is **share regression**, which returns to first-order conditions. Under competitive factor markets and profit maximization, the labor first-order condition implies that the observed labor expenditure share reveals the output elasticity of labor. In the Cobb-Douglas case,

$$\frac{w_t L_{jt}}{p_t Y_{jt}} = \frac{F_L(L_{jt}, K_{jt}) L_{jt} \mathbb{E}[e^{\eta_{jt}}]}{F(L_{jt}, K_{jt}) e^{\eta_{jt}}} = \beta_l \frac{\mathbb{E}[e^{\eta_{jt}}]}{e^{\eta_{jt}}}.$$

Taking logs gives

$$s_{jt} = \log(\beta_l) + \log \mathbb{E}[e^{\eta_{jt}}] - \eta_{jt}.$$

So the labor share provides direct identifying variation for β_l .

Its strength is that it avoids relying solely on proxy variation. Its limitation is that it requires strong structural assumptions about the factor market and the firm's optimization problem.

Problem 16. A synthesis question: choosing among OP, LP, ACF, and one-step GMM

Question. Suppose you have firm-level panel data on output, labor, capital, materials, investment, and firm exit. Investment is zero for many firms. Which estimator would you start from, and what additional issues would you check before treating the estimates as credible?

Detailed answer. A sensible starting point is usually **LP or a Wooldridge-style one-step version of LP**, because frequent zero investment makes the original OP inversion fragile. Materials are often a more reliable proxy for anticipated productivity than investment in that setting.

That said, I would not stop there. I would check four things.

- 1. Selection from exit.** If survival is nonrandom, I would include an OP-style correction by estimating the continuation probability and allowing the second-stage control function to depend on the propensity score.
- 2. Collinearity and timing.** If labor is likely chosen using the same information as the proxy input, I would worry about the ACF critique. Then I would either impose a timing structure that restores identification or use share-regression information to identify labor elasticity.
- 3. Gross output versus value added.** If materials enter output directly, I would be careful with any timing-based ACF fix because the gross-output version creates new identification problems for the materials coefficient.
- 4. Productivity dynamics.** I would assess whether a first-order Markov law is plausible or whether the productivity process appears to have higher-order dynamics or additional state variables such as demand shocks. If those dimensions matter, a scalar-unobservable proxy model may be too simple.

So the practical answer is not “pick one estimator and trust it.” It is “choose the proxy method that fits the data, then test whether its underlying timing, selection, and state-space assumptions are economically plausible.”

Part III. Cost Function Estimation

Problem 17. Cost duality and why it matters

Question. State the duality result emphasized in the chapter. Under what conditions is a cost function integrable, and why can it be attractive to estimate costs directly rather than begin from a production function?

Detailed answer. The duality result says that, once one adds a competitive factor-market environment and cost minimization, a production function and a cost function contain equivalent information. Given a production function $F(x)$, one can define the cost function

$$C(y, p) = \min_x p'x \quad \text{subject to } F(x) \geq y.$$

Conversely, if a candidate cost function satisfies the right regularity conditions, there exists a unique production function consistent with it.

The chapter lists the key conditions as:

- positivity for positive prices and output,
- homogeneity of degree one in input prices,
- monotonicity in prices and output,
- concavity in input prices.

When a cost function satisfies the conditions needed to come from a genuine production technology, it is called **integrable**.

Estimating costs directly can be attractive because closed-form cost specifications are often easier to work with than closed-form duals of production functions. In many applications, the cost side is also closer to the economic decision problem the researcher wants to study. If one accepts cost minimization and competitive factor markets, then identifying the cost function is effectively equivalent to identifying the underlying production technology.

Problem 18. The translog cost function and integrability restrictions

Question. Write the translog cost function used in the chapter and list the main integrability restrictions that must be imposed.

Detailed answer. The translog cost function is

$$\ln c = \alpha_0 + \alpha'_p \ln p + \alpha_y \ln y + \frac{1}{2}(\ln p)' B_{pp}(\ln p) + (\ln p)' \beta_{py} \ln y + \frac{1}{2} \beta_{yy} (\ln y)^2.$$

It is a second-order log-Taylor approximation to a general cost function and is popular because it allows flexible substitution patterns.

To ensure that the function is consistent with economic theory, the chapter emphasizes the following restrictions:

- **Homogeneity in prices.** The cost function must be homogeneous of degree one in input prices. This requires the coefficients to satisfy conditions such as

$$B_{pp}\mathbf{1} = 0, \quad \beta'_{py}\mathbf{1} = 0.$$

- **Cost exhaustion.** The cost shares must sum to one, implying

$$\alpha'_p\mathbf{1} = 1,$$

together with compatible zero-sum restrictions on higher-order terms.

- **Symmetry.** The matrix of second-order terms must be symmetric. Economically, this corresponds to the usual symmetry restrictions on substitution effects.
- **Monotonicity.** The implied cost shares and share elasticities must satisfy nonnegativity-type restrictions. The chapter expresses this through positive semidefiniteness of the relevant share-elasticity matrix.

So the translog is flexible, but its flexibility is useful only if these theory-consistent restrictions are enforced.

Problem 19. Cost-data versus revealed-preference approaches

Question. Compare the two broad approaches to cost estimation described in the chapter: the cost-data approach and the revealed-preference approach.

Detailed answer. The **cost-data approach** uses accounting or reported cost data directly and fits a cost function to those data. Its main advantage is that it can be relatively robust to strong behavioral assumptions. The researcher may impose cost-minimization restrictions if desired, but the approach does not mechanically require a fully specified product-market model.

Its weakness is measurement. Accounting costs do not always coincide with economic costs. They may miss adjustment costs, opportunity costs, or shadow costs that matter for actual firm decisions.

The **revealed-preference approach** instead infers cost parameters from firms' choices, typically prices and quantities, under an assumed model of competition and profit maximization. Its advantage is that it can recover the costs that firms behave *as if* they face, even when accounting data are incomplete or unavailable.

Its weakness is structural dependence. The estimated cost function is only as credible as the demand system, conduct assumptions, and equilibrium model used to recover marginal cost. If the market structure is misspecified, cost estimates will inherit that mistake.

A good short summary is this: cost-data methods lean on measured costs; revealed-preference methods lean on the behavioral model.

Problem 20. Single-product monopoly: identifying marginal cost from demand

Question. Suppose a single-product monopolist faces demand $D(p, z)$ and cost function $C(q)$. Derive the expression for marginal cost implied by the firm's first-order condition and explain why demand shifters are needed to recover the whole marginal cost curve.

Detailed answer. The monopolist solves

$$\max_p D(p, z)p - C(D(p, z)).$$

The first-order condition is

$$D(p, z) + pD_p(p, z) - C'(D(p, z))D_p(p, z) = 0.$$

Rearranging,

$$C'(D(p, z)) = \frac{D(p, z) + pD_p(p, z)}{D_p(p, z)}.$$

So if demand is known and price is observed, the first-order condition identifies the marginal cost at the equilibrium quantity. Intuitively, the firm's markup formula reveals the shadow value of output expansion.

Why are demand shifters z needed? Because a single equilibrium point only gives one point on the marginal-cost schedule. Exogenous shifts in demand move the firm along the marginal-cost curve by changing the equilibrium quantity. If those shifts generate enough variation in quantities, the researcher can trace out the entire function $C'(q)$.

Without excluded demand shifters, the problem is underidentified unless one imposes a very strong functional-form restriction such as constant marginal cost.

Problem 21. Multiproduct monopoly

Question. How does cost identification change when a monopolist sells multiple products? Give the matrix intuition from the chapter.

Detailed answer. With multiple products, the firm chooses the price vector p to maximize

$$\sum_{j=1}^J p_j D_j(p) - C(D_1(p), \dots, D_J(p)).$$

The first-order condition for price p_i is

$$D_i(p) + \sum_{j=1}^J p_j \frac{\partial D_j(p)}{\partial p_i} = \sum_{j=1}^J \frac{\partial C(q)}{\partial q_j} \frac{\partial D_j(p)}{\partial p_i}.$$

Stacking these conditions across products gives a linear system in the vector of marginal costs:

$$(\text{demand terms and price effects}) = (\text{Jacobian of demand}) \times (\text{vector of marginal costs}).$$

Therefore,

$$mc = [\text{Jacobian of demand}]^{-1} [\text{observed demand-price term}].$$

The economic intuition is that a multiproduct monopolist internalizes cannibalization across its own products. Raising the price of one product shifts demand toward or away from the firm's other products, and the firm takes those cross-effects into account. That is why the full demand Jacobian, not just the own-price derivative, matters for recovering marginal costs.

Problem 22. Oligopoly with one product per firm

Question. In the chapter's simple Bertrand setting, each firm sells one product. Derive the implied marginal cost expression and explain how this differs from the multiproduct monopoly case.

Detailed answer. Firm j solves

$$\max_{p_j} D_j(p)p_j - C_j(D_j(p)).$$

The first-order condition is

$$D_j(p) + \frac{\partial D_j(p)}{\partial p_j} p_j = \frac{\partial C_j(D_j(p))}{\partial q_j} \frac{\partial D_j(p)}{\partial p_j}.$$

Hence the marginal cost is

$$mc_j = \frac{D_j(p) + \frac{\partial D_j(p)}{\partial p_j} p_j}{\frac{\partial D_j(p)}{\partial p_j}}.$$

Relative to the multiproduct monopoly case, the difference is ownership. A single-product firm does not internalize the profit effects on rival products, so only its own demand derivative appears in the first-order condition. Cross-price effects still matter for demand, but they affect firm j only through the equilibrium price vector, not through an ownership link to other products.

This is why the same observed prices can imply different marginal costs under different assumptions about ownership and conduct.

Problem 23. Unobserved heterogeneity in the cost function

Question. Suppose the cost function is

$$C(q) = \tilde{C}(q) + q(z'\gamma + \epsilon) + \mu,$$

where ϵ and μ are unobserved. Explain how this complicates revealed-preference cost estimation and why excluded demand shifters become important.

Detailed answer. The complication is that the unobserved cost component enters firms' choices. In the single-product monopoly example, marginal cost becomes

$$MC(q) = \tilde{C}'(q) + z'\gamma + \epsilon.$$

Because ϵ is anticipated by the firm, it affects pricing and therefore equilibrium output. That makes it endogenous from the econometrician's point of view.

A common strategy is to proceed in two steps:

- Use the pricing first-order condition and an estimated demand system to back out an implied marginal cost:

$$mc = \frac{D(p, x) + pD_p(p, x)}{D_p(p, x)} - \tilde{C}'(D(p, x)).$$

- Regress the part of marginal cost associated with observed cost shifters on z :

$$mc = z'\gamma + \epsilon,$$

using moment conditions such as $E[\epsilon \mid x] = 0$, where x are demand shifters excluded from the cost function.

The reason excluded demand shifters matter is that the observed cost shifters z are already being used to identify the linear part of marginal cost. To identify the nonlinear term $\tilde{C}'(q)$, the researcher needs extra variation in demand that moves quantity without directly moving costs. If one assumes constant marginal cost so that $\tilde{C}'(q) = 0$, the need for such demand shifters becomes much weaker, which is why that assumption is so common in practice.

Problem 24. A final big-picture comparison

Question. Compare production-function estimation and cost-function estimation as empirical strategies. When would you prefer one over the other?

Detailed answer. Production-function estimation is the natural approach when the researcher's core question is about **technology and productivity**. It is especially attractive when the goal is to understand how policy, trade, management, R&D, or distortions affect firms' underlying ability to transform inputs into output.

Cost-function estimation is natural when the researcher's core question is about **costs as they matter for behavior**. Through duality, cost functions can still reveal technology, but they are often closer to pricing and market-conduct questions. In IO, that is particularly useful when one ultimately wants marginal costs for merger simulation, markup measurement, or pass-through analysis.

In practice, the choice depends on both the economic question and the data:

- If you have rich plant-level input and output data and care about firm productivity, production-function methods are often the first choice.
- If you have good pricing and demand information but poor direct cost data, revealed-preference cost methods can be more informative.
- If accounting cost data are available and reasonably reliable, direct cost estimation can be simpler and less dependent on market-structure assumptions.

So the two approaches are complements, not substitutes. The correct starting point is the economic question you want to answer and the assumptions you are willing to defend.